

JAWAHARLAL NEHRU UNIVERSITY

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Minor Exam

M.CA. (Winter Semester-2024)

Course Name: Computer Architecture

Course Code: CS 106

Maximum Marks: 30

Total Time: 1 hour

Q1. What is the difference between RISC and CISC type of Computer Architecture?.

A computing task is divided into three parts:

1. Part A: 50% of the task can be fully parallelized.
2. Part B: 30% of the task can be parallelized.
3. Part C: 20% of the task is inherently serial and cannot be parallelized.

Calculate the overall speedup when using 8 processors for Part A and 4 processors for Part B. Part C remains serial.

(4+6)

Q2. (a) What are the major performance parameters that determines the execution time of processor?.

b) What is Harvard Architecture?.

- (a) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. ii) reduce CPI instruction type C to 1.8. Which one is better?.

Type	CPI	Frequency
A	1.3	40%
B	2.3	40%
C	2.1	20%

(5+2+3)

Q3. (a) What are the different types of addressing modes that in 8086 microprocessor?.

- (b) Write an assembly language program to subtract two numbers using 2^{nd} complement. What is the new memory capacity if the RAM memory is scaled upto from 2^{20} byte to 2^{28} byte?.

(4+4+2)

JAWAHARLAL NEHRU UNIVERSITY

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Mid-Term Exam

M.CA. (Winter Semester-2024)

Course Name: Computer Architecture

Maximum Marks: 45

Course Code: CS 106

Total Time: 2 hour

Q1. a) Which among Harvard and Von Neumann Architecture is preferred and why?.

Three enhancements with following speedups are proposed for new architecture:

Speedup₁=30; Speedup₂=20; Speedup₃=15; only one enhancement is useable at time

(a) If enhancement 1 and 2 are each usable for 18% of the time, what fraction of the time must enhancement 3 be used to achieve overall speed up of 18?.

(b) Assume the enhancement can be used 30%, 40% and 30% of the time for enhancement 1, 2, 3 respectively. For what fraction of the reduced execution time no enhancement is used?.

(4+6)

Q2. (a) Which performance parameters can improve the execution time of the processor by improving the Computer Architecture?.

b) A weather forecasting computation requires 400 billion floating point operations. The problem is processed in a Super computer that can perform 80 megaflops. How long will it take to perform these calculations?

c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 2.1. Which one is worse?.

Type	CPI	Frequency
A	1.5	40%
B	2.5	40%
C	2.0	20%

(3+3+4)

Q3. (a) Why there is a need for Memory Hierarchy in Computer Architecture. Explain with important parameters the Memory Hierarchy in Computer System?. How can one increase the relative size in a Memory system?.

b) What is the difference between SRAM and DRAM cell. Explain with their working.

(5+5)

Q4. (a) What are the total external connections in 256X16 Memory system?. What is the 90/10 rule?. What is the difference between Temporal and Spatial locality in memory system?.

(b) Explain the difference between Direct, Full Associative and Set Associative cache mapping. Which one is preferred and why ?.

(5+3+2)

Q5. (a) Consider a CPU with an average CPI as 1.8. Assume the instruction mix is ALU-35%, Load-15%, Store-20% Branch-30%. Assume the cache miss rate of 5% and miss penalty of 60 cycles ($=t_{MM}$). Calculate the effective CPI for a unified L1 cache.

(5)

JAWAHARLAL NEHRU UNIVERSITY

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Final Exam

M.CA. (Winter Semester-2024)

Course Name: Computer Architecture

Maximum Marks: 60

Course Code: CS 106

Total Time: 3.0 hours

- Q1.** (a) Why RISC Architecture is preferred over X86 in the present day smart phones?
 (b) What are the factors affecting the performance of Computer Architecture
 (c) Determine the normalized execution time using arithmetic mean and the geometric mean

Type	CPU A	CPU B	CPU C
Program P1	5	15	25
Program P2	250	100	15
Total Time			

Which one is preferred and why

(3+3+4)

- Q2.** (a) Explain the concept of parallelism or pipelining in context to CPU and GPU.
 (b) A quad core processor could speed up a computer by a factor of 4 but this rarely happens. Compute the percentage of program execution that needs to be distributed across all 4 cores to achieve an overall speedup of 3. Of 2. Of 1.5. Of 1.25.

(c) You have a computing task that can be divided into three parts:

1. Part A: 50% of the task can be fully parallelized.
2. Part B: 30% of the task can be parallelized.
3. Part C: 20% of the task is inherently serial and cannot be parallelized.

You want to calculate the overall speedup when using 16 processors for Part A and 8 processors for Part B. Part C remains serial.

(3+4+5)

Q3. (a) What is the difference between cache memory, flash memory and virtual? Explain the term locality of reference and spatial locality.

(b) Explain about the type of memory used in primary memories.

(c) Write an assembly program to explain different types of addressing modes in 8085 microprocessor.

(5+2+3)

Q4. (a) What are the basic cache optimization techniques?

(b) Calculate the average access time in both types of memory hierarchy for the following parameters:

- i) Clock cycle is 5 ns
- ii) Miss cycle is 10 clock cycle
- iii) 5 percent of instruction is not found in Instruction cache
- iv) 10 percent of instruction is not found in Data cache
- v) 25 percent of the total memory access for the data
- vi) Cache Access Time (including hit detection) is 1 clock cycle.

Which one of the memory configuration is better?

(3+6)

Q5. (a) What is Dynamic and a Non-linear pipelining technique?

(b) For the Multiplier and Addition reservation table

- (i) Determine all the forbidden latencies
- (ii) Show the state transition diagram
- (iii) Determine the optimum MAL
- (v) Determine the pipeline throughput with $\Delta = 20\text{ns}$
- (vi) What is the efficiency of the pipeline technique.

(3+7)

Q6. (a) Explain the features of vector processor Architecture?. What are the different Taxonomy of Parallel Architectures.

(b) What is the advantage of using multi core?. Explain the difference between loosely coupled and tightly coupled Microprocessor.

(4+4)